MILITARY SPECIFICATION

DRILL, HAND, BONE, SMEDBERG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a drill bone brace, hand type, of the Smedberg design, suitable for use in orthopedic surgery.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping Fiberboard.
PPP-B-676	Box, Set-up, Paperboard

Military

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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DRAWING

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material and hardness requirements shall be as specified in Table I. The hardness requirements of all class 4 material shall be after proper stress relieving. All other material not indicated shall be fabricated from corrosion-resistant metal and suitable for the purpose intended.

Table I. Material and Hardness

Component	Class	Type	Rockwell Hardness	Specification
Body	-	Aluminum alloy	-	-
Drill shank	3 or 4	302, 303, 304 or 416	B70 minimum for class 3; C36 to C41 for class 4	GG-I-526 American Iron Steel Institute (AISI)
	or	**AISI 440F	C36 to C41	
Operating handle	8	260, 270 or 360	-	GG-I-526
	or	Sand Casting copper alloy 836 chromium plated over nickel plated	-	Chemical composition in accordance with ASTM B 145
Jacobs Chuck	4	416	C36 to C41	GG-I-526
	or	**AISI 440F	C36 to C41	AISI
Thumb locking lever	8	260, 270 or 360	-	GG-I-526
	or	Sand casting copper alloy 836 chromium plated over nickel plated	-	Chemical composition in accordance with ASTM B 145
* Chuck Key	4	416	C35 to C41	GG-I-526
	or	SAE 1119 or SAE 1117 Carbon steel	C50 to C60	Society of Automotive Engineers Inc. (SAE)

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Table I. Material and Hardness (cont'd)

Component	Class	Type	Rockwell Hardness	Specification
Chuck Key gear	4 or	416 SAE 1119 or SAE 1117 Carbon steel	C36 to C41 C50 to C60	GG-I-526 SAE
Chuck Key handle	4 or	416 SAE 8640 or SAE 8650	C36 to C41 C50 to C60	GG-I-526 SAE
Operating bevel gears	-	Copper alloy	1/2 hard rod	SAE alloy 72 (reference)

*When entire chuck key (key, gear and handle) is fabricated from carbon steel alloy, it shall be chromium plated over nickel plated. The corrosion tests shall be the salt spray and boil tests, and the plating requirements shall comply with class 1, type 1095 material of GG-I-526.

**AISI 440F material shall comply with corrosion resistance requirements for class 4, type 416 material of GG-I-526.

3.2 Passivation. All material fabricated from 302, 303, 304 or 416 or AISI 440F material shall be properly passivated.

3.3 Style, design and dimensions. Shall be a bone drill brace, universal, hand type, of the Smedberg design, suitable for use in orthopedic surgery. Shall consist essentially of a one-piece body with removable gear container cover, No. "0" Jacobs type chuck, pistol grip formed handle, Allen wrench type handle chuck key, and thumb operated locking lever. Assembly of chuck to drill shank shall be by means of suitable mating tapers as indicated in drawing. The driving mechanism shall consist of two bevel gears; one secured to the drill shank and the other gear secured to the operating handle. The gears shall be designed in such fashion that one complete revolution of the operating handle shall result in two complete revolutions of the chuck. The thumb locking lever shall be of the cam-locking design that shall prevent turning of the operating handle when the locking lever is not depressed. The entire assembly shall be suitable for use and accept Steinmann type bone pins, bayonet and trocar points, triangular shank, sizes 3/32 to 5/32 inch in diameter, 4 to 12 inches in overall length; bone twist drills sizes 3/32 to 5/32 inch in diameter; and Kirschner bone pins sizes .035 to .062 inch in diameter by 9 to 12 inches in overall length. The drill assembly shall be cannulated throughout its length to accept drills up to 12 inches in overall length.

The No. "0" Jacobs chuck shall have a standard No. 0 taper, as indicated on drawing, with unthreaded tapered hole. The chuck shall accept bone pins and drills having diameters up to, and including, 5/32 inch.

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The chuck key shall be of the universal type suitable for use with the No. "0" Jacobs chuck. The key handle shall be of the Allen wrench-type suitable for use with number 10 socket head screws. The key gear shall mesh easily with the chuck gear without slipping. The complete drill assembly shall be in accordance with DPSC Drawing No. 20488.

3.4 Finish. Finish of the drill body, Jacobs chuck and chuck gear key shall be mirror, or mirror and satin.

3.5 Identification marking. The drill proper and chuck shall be permanently marked with the name or registered trade mark of the contractor, and the country of origin, if other than the U.S.

3.6 Service data. The contractor shall enclose, with each drill assembly two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Personnel Support Center, Directorate of Medical Materiel. The cover of each service manual should indicate the following:

NSN 6515-00-312-3500
Drill, Hand, Bone, Smedberg
Contractor's trademark, Model
Number, Name and Address

The preliminary service data shall be submitted sufficiently early to permit adequate review by the Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

3.7 Workmanship. Workmanship shall be first class throughout. Drill assembly shall be free from burrs, cracks, pits, sharp edges, scale, corrosion and other defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except

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as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel alloys and copper alloys, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. The unit of product for examination purposes shall be one complete drill assembly or one Jacobs chuck, or one key, as applicable (see 6.2).

Table II. Sampling Criteria for Examination

Type defect	Inspection level	AQL (percent defective
Major	II	1.0
Minor	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table III. The unit of product for test purposes shall be one complete drill assembly, or one chuck, or one key, as applicable (see 6.2).

Table III. Sampling Criteria for Tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Chuck & Drill shank	Hardness	3.1	GG-I-526	S-2	1.5
	Copper Sulfate test	3.1 & GG-I-526	GG-I-526	S-2	1.0
	Boil test	3.1 & GG-I-526	GG-I-526	S-2	1.0
Chuck key	Hardness	3.1	GG-I-526	S-2	1.5
	Plating test (carbon steel)	3.1 & GG-I-526	GG-I-526	S-2	1.0
	Salt-spray test (carbon steel)	3.1 & GG-526	GG-I-526	S-2	1.0
	Boil test (all material)	3.1 & GG-526	GG-I-526	S-2	1.0
	Copper sulfate test (class 4 material)	3.1 & GG-I-526	GG-I-526	S-2	1.0
Operating handle and thumb locking lever	Plating test (all material)	3.1 & GG-I-526	GG-I-526	S-2	1.0
	Salt-spray test	3.1 & GG-I-526	GG-I-526	S-2	1.0
	Boil test	3.1 & GG-I-526	GG-I-526	S-2	1.0
Drill assembly	Dimensions	3.3	---	S-2	2.5

4.2.3 Raw materials. A sufficient quantity of each lot of each metal used in the construction of the hand drills (or chuck and key when procured separately) shall be selected for chemical analysis.

4.2.4 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination of assemblies. The hand drill assemblies (or chuck and key when procured separately) shall be examined to determine compliance with all requirements contined in this specification. Examination shall be

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conducted as specified in Tables II and IV. Examination, however, shall not be restricted to defects listed in Table IV. Any items in the sample containing one or more defects shall be rejected and if the number of defective hand drills in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table IV. Classification of Defects.

Categories	Defects
Critical	None defined.
Major	
101	Drill assembly (including chuck and key) not of the design specified.
102	Drill not cannulated throughout to accept bone pins and drills specified.
103	Drill excessively stiff and not smooth in operation when handle is turned.
104	Locking lever does not function.
105	Components not free from cracks, pits, burrs, sharp edges, scale and corrosion.
106	Drill not furnished with all specified components.
107	Plated parts not free from flaking or not plated uniformly.
108	Chuck does not hold pins and drills up to 5/32 inch diameter tightly and securely.
109	Key gear does not mesh easily with chuck gear without slipping.
110	One complete turn of handle does not result in two complete revolutions of chuck.
111	Finish not as specified.
Minor	
201	Identification markings not legible, not complete, not correct, or missing.
202	Service data not furnished.

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4.3.1 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table V. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table V. Examination for Preparation for Delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packing and packaging materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each drill assembly or each chuck or each key shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.1.2 Preservative. Preservative shall not be applied.

5.1.1.3 Unit package. Each drill assembly, or each chuck or each key shall be wrapped in neutral tissue or other suitable anticorrosive materials. Drill assembly or chuck shall be further packaged in accordance with MIL-P-116, method 1C-2. Key shall be further packaged in accordance with MIL-P-116, method 1C-1.

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5.1.1.4 Intermediate package. Twelve (12) chucks or twelve (12) keys shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-676 or PPP-B-636, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twelve (12) drill assemblies or two hundred and eightly eight (288) chucks or five hundred and seventy six (576) keys shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseass type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard boxes shall be waterproof in accordance with PPP-B-636.

5.2.2 Level B. Twelve (12) drill assemblies or two hundred and eighty eight (288) chucks or five hundred and seventy six (576) keys shall be packed in a box designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit and each intermediate package shall be marked in accordance with MIL-STD-129.

5.3.2 Exterior container. Each exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement document should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number of complete drill assembly, or chuck, or key, as applicable (see 6.2).
(c) Applicable levels of packing and packaging (see 5.1 and 5.2).

6.2 This specification covers the following items appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-312-3500	DRILL, HAND, BONE, SMEDBERG
6515-00-312-3510	CHUCK, DRILL, Jacobs, Capacity 0 to 5/32 inches.
6515-00-312-3530	KEY, DRILL CHUCK

6,3 Supersession data. This specification includes requirements for purchase description of items respectively listed in 6.2 as indicated below:

Purchase Description No. 9, dated 21 May 1969;
Purchase Description No. 5, dated 26 August 1968;
Purchase Description No. 6, dated 24 October 1968.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1933
CS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-D-37215 DRILL, HAND, BONE, CMEDBERG

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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